

A Normal-reference data

NSRDB: ECG1

18 complete recordings
Use the first stored channel only.



Clean waveform windows

512-sample, non-overlapping windows
Retain archived amplitude scaling
288,673 fitting windows
57,177 validation windows



Train, validate, then freeze

15 subjects fit the autoencoder.
3 other subjects select its checkpoint.
Apply it to each MLII beat to obtain
4 reconstruction-error features.



B Labelled MLII beat data

MITDB: MLII

46 eligible records
Resample to 128 Hz; locate R peaks.



Prepare one accepted beat

Extract 512 samples around the R peak.
Filter and z-score within that window.
Measure previous and next RR intervals
and earlier rhythm history.



One feature vector per beat

223 waveform measurements
+ 15 RR timing measurements
+ 4 autoencoder error measurements
= 242 values for both tree models

Expert labels are stored separately from the 242 input features.

Earlier MITDB beats develop the models; later beats evaluate their predictions.
The first and last eligible beat are excluded when a neighbouring RR is unavailable.